Experimental Design Workflow Library

Expert Workflow Library

Ultimate AI Research Toolkit 2026 - Premium Edition

Designed for PhD students, professors, graduate researchers, engineers, and data scientists.

[Idea]	Research prompts and ideation systems
[Review]	Literature synthesis and gap detection
[Write]	Paper writing and revision workflows
[Plan]	Publication and productivity operations

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Introduction

This Experimental Design collection provides 100 expert research workflows engineered for high-rigor academic and technical delivery. Each workflow includes practical fields to help users move from strategy to publication-ready output.

Instructions

- Select workflows by category and difficulty to match project maturity.
- Replace variables in each workflow with your discipline, data, and resource constraints.
- Use Expected Output as the acceptance criteria for quality control.
- Store outputs in versioned folders to preserve decision history.

Workflow Library

1. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - practical deployment framing

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize rigor and produce publication-grade decisions for Control Groups.

Prompt: Rewrite the methodology around Control Groups for replication package for supplementary material so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ICML. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=benchmarking specialist. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Strategy

2. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - method comparison clarity

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize reproducibility and produce publication-grade decisions for Evaluation Protocol.

Prompt: Write a Nature-style abstract for a study on Evaluation Protocol in high-variance real-world deployment. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=theory-oriented reviewer. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Analysis

3. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - editorial risk reduction

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize clarity and produce publication-grade decisions for Bias Mitigation.

Prompt: Generate a figure strategy for Bias Mitigation with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by IEEE TPAMI. Add one caption rewrite for non-expert readers. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=methodology purist. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Writing

4. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - venue-specific optimization

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize statistical power and produce publication-grade decisions for Ablation Planning.

Prompt: Build an ethics checklist and mitigation plan for Ablation Planning in multi-dataset benchmarking study. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for NeurIPS review. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=interdisciplinary committee member. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Publication

5. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - decision traceability

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize transferability and produce publication-grade decisions for Control Groups.

Prompt: Design a publication strategy for Control Groups: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ICML. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=ethics compliance reviewer. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Methodology

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6. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - citation-strength positioning

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize ethical compliance and produce publication-grade decisions for Evaluation Protocol.

Prompt: Assume Reviewer #2 challenges Evaluation Protocol for AAAI. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=reproducibility auditor. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Execution

7. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - defense-ready reasoning

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize computational efficiency and produce publication-grade decisions for Bias Mitigation.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Bias Mitigation. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for IEEE TPAMI. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=applied domain expert. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Compliance

8. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - high-confidence statistical reporting

Objective: Select robust statistical strategy and interpretation logic. Prioritize clinical relevance and produce publication-grade decisions for Ablation Planning.

Prompt: Design a statistical analysis workflow for Ablation Planning with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for NeurIPS. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=systems performance reviewer. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Reasoning

9. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - evidence-first writing

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize publication readiness and produce publication-grade decisions for Control Groups.

Prompt: Analyze a draft on Control Groups and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ICML. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=industry impact reviewer. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Analysis

10. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - hypothesis stress testing

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize novelty and produce publication-grade decisions for Evaluation Protocol.

Prompt: Interpret results for Evaluation Protocol using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in AAAI. Add one paragraph that preempts overclaiming. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=editor focused on clarity. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Writing

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11. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - results-to-insight translation

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize rigor and produce publication-grade decisions for Bias Mitigation.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Bias Mitigation and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for IEEE TPAMI. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=skeptical statistician. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Publication

12. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - major-revision readiness

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize reproducibility and produce publication-grade decisions for Ablation Planning.

Prompt: Generate a publication-ready section for Ablation Planning tuned for NeurIPS. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=benchmarking specialist. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Methodology

13. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - replication-grade documentation

Objective: Pressure-test experimental design before expensive execution. Prioritize clarity and produce publication-grade decisions for Control Groups.

Prompt: Audit the experimental design for Control Groups. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ICML reviewers. Include a stop/go decision protocol. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=theory-oriented reviewer. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Execution

14. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - reviewer-resilient argumentation

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize statistical power and produce publication-grade decisions for Evaluation Protocol.

Prompt: Create a LaTeX writing workflow for Evaluation Protocol: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for AAAI submission checks. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=methodology purist. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Compliance

15. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - ethics-by-design reporting

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize transferability and produce publication-grade decisions for Bias Mitigation.

Prompt: Simulate a high-pressure Q&A; on Bias Mitigation. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=interdisciplinary committee member. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Reasoning

16. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - limitations-first transparency

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize ethical compliance and produce publication-grade decisions for Ablation Planning.

Prompt: Act as a senior research editor. Build a contribution architecture for Ablation Planning in multi-dataset benchmarking study targeting NeurIPS. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=ethics compliance reviewer. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Rebuttal

17. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - cross-domain readability

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize computational efficiency and produce publication-grade decisions for Control Groups.

Prompt: Rewrite the methodology around Control Groups for replication package for supplementary material so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ICML. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=reproducibility auditor. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Validation

18. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - impact-focused communication

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize clinical relevance and produce publication-grade decisions for Evaluation Protocol.

Prompt: Write a Nature-style abstract for a study on Evaluation Protocol in high-variance real-world deployment. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=applied domain expert. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Publication

19. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - publishable novelty framing

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize publication readiness and produce publication-grade decisions for Bias Mitigation.

Prompt: Generate a figure strategy for Bias Mitigation with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by IEEE TPAMI. Add one caption rewrite for non-expert readers. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=systems performance reviewer. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Methodology

20. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - practical deployment framing

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize novelty and produce publication-grade decisions for Ablation Planning.

Prompt: Build an ethics checklist and mitigation plan for Ablation Planning in multi-dataset benchmarking study. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for NeurIPS review. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=industry impact reviewer. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Execution

21. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - method comparison clarity

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize rigor and produce publication-grade decisions for Control Groups.

Prompt: Design a publication strategy for Control Groups: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ICML. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=editor focused on clarity. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Compliance

22. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - editorial risk reduction

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize reproducibility and produce publication-grade decisions for Evaluation Protocol.

Prompt: Assume Reviewer #2 challenges Evaluation Protocol for AAAI. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile:

discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=skeptical statistician. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Reasoning

23. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - venue-specific optimization

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize clarity and produce publication-grade decisions for Bias Mitigation.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Bias Mitigation. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for IEEE TPAMI. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=benchmarking specialist. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Rebuttal

24. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - decision traceability

Objective: Select robust statistical strategy and interpretation logic. Prioritize statistical power and produce publication-grade decisions for Ablation Planning.

Prompt: Design a statistical analysis workflow for Ablation Planning with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for NeurIPS. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline};

resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=theory-oriented reviewer. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Validation

25. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - citation-strength positioning

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize transferability and produce publication-grade decisions for Control Groups.

Prompt: Analyze a draft on Control Groups and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ICML. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=methodology purist. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Optimization

26. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - defense-ready reasoning

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize ethical compliance and produce publication-grade decisions for Evaluation Protocol.

Prompt: Interpret results for Evaluation Protocol using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in AAAI. Add one paragraph that preempts overclaiming. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=interdisciplinary committee member.

Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Planning

27. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - high-confidence statistical reporting

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize computational efficiency and produce publication-grade decisions for Bias Mitigation.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Bias Mitigation and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for IEEE TPAMI. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=ethics compliance reviewer. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Execution

28. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - evidence-first writing

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize clinical relevance and produce publication-grade decisions for Ablation Planning.

Prompt: Generate a publication-ready section for Ablation Planning tuned for NeurIPS. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=reproducibility auditor. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Compliance

29. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - hypothesis stress testing

Objective: Pressure-test experimental design before expensive execution. Prioritize publication readiness and produce publication-grade decisions for Control Groups.

Prompt: Audit the experimental design for Control Groups. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ICML reviewers. Include a stop/go decision protocol. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=applied domain expert. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Reasoning

30. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - results-to-insight translation

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize novelty and produce publication-grade decisions for Evaluation Protocol.

Prompt: Create a LaTeX writing workflow for Evaluation Protocol: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for AAAI submission checks. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=systems performance reviewer. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Rebuttal

31. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - major-revision readiness

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize rigor and produce publication-grade decisions for Bias Mitigation.

Prompt: Simulate a high-pressure Q&A; on Bias Mitigation. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=industry impact reviewer. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Validation

32. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - replication-grade documentation

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize reproducibility and produce publication-grade decisions for Ablation Planning.

Prompt: Act as a senior research editor. Build a contribution architecture for Ablation Planning in multi-dataset benchmarking study targeting NeurIPS. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=editor focused on clarity. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Optimization

33. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - reviewer-resilient argumentation

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize clarity and produce publication-grade decisions for Control Groups.

Prompt: Rewrite the methodology around Control Groups for replication package for supplementary material so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ICML. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=skeptical statistician. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Planning

34. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - ethics-by-design reporting

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize statistical power and produce publication-grade decisions for Evaluation Protocol.

Prompt: Write a Nature-style abstract for a study on Evaluation Protocol in high-variance real-world deployment. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=benchmarking specialist. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Strategy

35. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - limitations-first transparency

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize transferability and produce publication-grade decisions for Bias Mitigation.

Prompt: Generate a figure strategy for Bias Mitigation with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by IEEE TPAMI. Add one caption rewrite for non-expert readers. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=theory-oriented reviewer. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Analysis

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36. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - cross-domain readability

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize ethical compliance and produce publication-grade decisions for Ablation Planning.

Prompt: Build an ethics checklist and mitigation plan for Ablation Planning in multi-dataset benchmarking study. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for NeurIPS review. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=methodology purist. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Reasoning

37. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - impact-focused communication

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize computational efficiency and produce publication-grade decisions for Control Groups.

Prompt: Design a publication strategy for Control Groups: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ICML. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=interdisciplinary committee member. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Rebuttal

38. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - publishable novelty framing

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize clinical relevance and produce publication-grade decisions for Evaluation Protocol.

Prompt: Assume Reviewer #2 challenges Evaluation Protocol for AAAI. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=ethics compliance reviewer. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Validation

39. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - practical deployment framing

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize publication readiness and produce publication-grade decisions for Bias Mitigation.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Bias Mitigation. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for IEEE TPAMI. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=reproducibility auditor. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Optimization

40. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - method comparison clarity

Objective: Select robust statistical strategy and interpretation logic. Prioritize novelty and produce publication-grade decisions for Ablation Planning.

Prompt: Design a statistical analysis workflow for Ablation Planning with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for NeurIPS. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=applied domain expert. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Planning

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41. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - editorial risk reduction

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize rigor and produce publication-grade decisions for Control Groups.

Prompt: Analyze a draft on Control Groups and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ICML. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=systems performance reviewer. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Strategy

42. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - venue-specific optimization

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize reproducibility and produce publication-grade decisions for Evaluation Protocol.

Prompt: Interpret results for Evaluation Protocol using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in AAAI. Add one paragraph that preempts overclaiming. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=industry impact reviewer. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Analysis

43. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - decision traceability

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize clarity and produce publication-grade decisions for Bias Mitigation.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Bias Mitigation and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for IEEE TPAMI. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=editor focused on clarity. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Writing

44. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - citation-strength positioning

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize statistical power and produce publication-grade decisions for Ablation Planning.

Prompt: Generate a publication-ready section for Ablation Planning tuned for NeurIPS. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=skeptical statistician. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Publication

45. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - defense-ready reasoning

Objective: Pressure-test experimental design before expensive execution. Prioritize transferability and produce publication-grade decisions for Control Groups.

Prompt: Audit the experimental design for Control Groups. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ICML reviewers. Include a stop/go decision protocol. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=benchmarking specialist. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Validation

46. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - high-confidence statistical reporting

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize ethical compliance and produce publication-grade decisions for Evaluation Protocol.

Prompt: Create a LaTeX writing workflow for Evaluation Protocol: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for AAAI submission checks. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=theory-oriented reviewer. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Optimization

47. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - evidence-first writing

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize computational efficiency and produce publication-grade decisions for Bias Mitigation.

Prompt: Simulate a high-pressure Q&A; on Bias Mitigation. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=methodology purist. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Planning

48. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - hypothesis stress testing

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize clinical relevance and produce publication-grade decisions for Ablation Planning.

Prompt: Act as a senior research editor. Build a contribution architecture for Ablation Planning in multi-dataset benchmarking study targeting NeurIPS. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=interdisciplinary committee member. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Strategy

49. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - results-to-insight translation

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize publication readiness and produce publication-grade decisions for Control Groups.

Prompt: Rewrite the methodology around Control Groups for replication package for supplementary material so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ICML. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=ethics compliance reviewer. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Analysis

50. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - major-revision readiness

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize novelty and produce publication-grade decisions for Evaluation Protocol.

Prompt: Write a Nature-style abstract for a study on Evaluation Protocol in high-variance real-world deployment. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=reproducibility auditor. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Writing

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51. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - replication-grade documentation

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize rigor and produce publication-grade decisions for Bias Mitigation.

Prompt: Generate a figure strategy for Bias Mitigation with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by IEEE TPAMI. Add one caption rewrite for non-expert readers. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=applied domain expert. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Publication

52. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - reviewer-resilient argumentation

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize reproducibility and produce publication-grade decisions for Ablation Planning.

Prompt: Build an ethics checklist and mitigation plan for Ablation Planning in multi-dataset benchmarking study. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for NeurIPS review. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=systems performance reviewer. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Methodology

53. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - ethics-by-design reporting

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize clarity and produce publication-grade decisions for Control Groups.

Prompt: Design a publication strategy for Control Groups: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ICML. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=industry impact reviewer. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Execution

54. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - limitations-first transparency

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize statistical power and produce publication-grade decisions for Evaluation Protocol.

Prompt: Assume Reviewer #2 challenges Evaluation Protocol for AAAI. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=editor focused on clarity. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Planning

55. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - cross-domain readability

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize transferability and produce publication-grade decisions for Bias Mitigation.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Bias Mitigation. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for IEEE TPAMI. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=skeptical statistician. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Strategy

56. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - impact-focused communication

Objective: Select robust statistical strategy and interpretation logic. Prioritize ethical compliance and produce publication-grade decisions for Ablation Planning.

Prompt: Design a statistical analysis workflow for Ablation Planning with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for NeurIPS. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=benchmarking specialist. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Analysis

57. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - publishable novelty framing

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize computational efficiency and produce publication-grade decisions for Control Groups.

Prompt: Analyze a draft on Control Groups and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ICML. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=theory-oriented reviewer. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Writing

58. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - practical deployment framing

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize clinical relevance and produce publication-grade decisions for Evaluation Protocol.

Prompt: Interpret results for Evaluation Protocol using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in AAAI. Add one paragraph that preempts overclaiming. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=methodology purist. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Publication

59. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - method comparison clarity

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize publication readiness and produce publication-grade decisions for Bias Mitigation.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Bias Mitigation and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for IEEE TPAMI. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=interdisciplinary committee member. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Methodology

60. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - editorial risk reduction

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize novelty and produce publication-grade decisions for Ablation Planning.

Prompt: Generate a publication-ready section for Ablation Planning tuned for NeurIPS. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=ethics compliance reviewer. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Execution

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61. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - venue-specific optimization

Objective: Pressure-test experimental design before expensive execution. Prioritize rigor and produce publication-grade decisions for Control Groups.

Prompt: Audit the experimental design for Control Groups. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ICML reviewers. Include a stop/go decision protocol. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=reproducibility auditor. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Compliance

62. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - decision traceability

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize reproducibility and produce publication-grade decisions for Evaluation Protocol.

Prompt: Create a LaTeX writing workflow for Evaluation Protocol: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for AAAI submission checks. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=applied domain expert. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Reasoning

63. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - citation-strength positioning

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize clarity and produce publication-grade decisions for Bias Mitigation.

Prompt: Simulate a high-pressure Q&A; on Bias Mitigation. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=systems performance reviewer. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Analysis

64. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - defense-ready reasoning

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize statistical power and produce publication-grade decisions for Ablation Planning.

Prompt: Act as a senior research editor. Build a contribution architecture for Ablation Planning in multi-dataset benchmarking study targeting NeurIPS. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=industry impact reviewer. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Writing

65. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - high-confidence statistical

reporting

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize transferability and produce publication-grade decisions for Control Groups.

Prompt: Rewrite the methodology around Control Groups for replication package for supplementary material so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ICML. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=editor focused on clarity. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Publication

66. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - evidence-first writing

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize ethical compliance and produce publication-grade decisions for Evaluation Protocol.

Prompt: Write a Nature-style abstract for a study on Evaluation Protocol in high-variance real-world deployment. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=skeptical statistician. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Methodology

67. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - hypothesis stress testing

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize computational efficiency and produce publication-grade decisions for Bias Mitigation.

Prompt: Generate a figure strategy for Bias Mitigation with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by IEEE TPAMI. Add one caption rewrite for non-expert readers. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=benchmarking specialist. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Execution

68. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - results-to-insight translation

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize clinical relevance and produce publication-grade decisions for Ablation Planning.

Prompt: Build an ethics checklist and mitigation plan for Ablation Planning in multi-dataset benchmarking study. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for NeurIPS review. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=theory-oriented reviewer. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Compliance

69. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - major-revision readiness

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize publication readiness and produce publication-grade decisions for Control Groups.

Prompt: Design a publication strategy for Control Groups: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ICML. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=methodology purist. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Reasoning

70. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - replication-grade documentation

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize novelty and produce publication-grade decisions for Evaluation Protocol.

Prompt: Assume Reviewer #2 challenges Evaluation Protocol for AAAI. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=interdisciplinary committee member. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Rebuttal

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71. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - reviewer-resilient argumentation

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize rigor and produce publication-grade decisions for Bias Mitigation.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Bias Mitigation. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for IEEE TPAMI. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=ethics compliance reviewer. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Validation

72. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - ethics-by-design reporting

Objective: Select robust statistical strategy and interpretation logic. Prioritize reproducibility and produce publication-grade decisions for Ablation Planning.

Prompt: Design a statistical analysis workflow for Ablation Planning with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for NeurIPS. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=reproducibility auditor. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Publication

73. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - limitations-first transparency

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize clarity and produce publication-grade decisions for Control Groups.

Prompt: Analyze a draft on Control Groups and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ICML. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=applied domain expert. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Methodology

74. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - cross-domain readability

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize statistical power and produce publication-grade decisions for Evaluation Protocol.

Prompt: Interpret results for Evaluation Protocol using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in AAAI. Add one paragraph that preempts overclaiming. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=systems performance reviewer. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Execution

75. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - impact-focused communication

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize transferability and produce publication-grade decisions for Bias Mitigation.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Bias Mitigation and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for IEEE TPAMI. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=industry impact reviewer. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Compliance

76. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - publishable novelty framing

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize ethical compliance and produce publication-grade decisions for Ablation Planning.

Prompt: Generate a publication-ready section for Ablation Planning tuned for NeurIPS. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=editor focused on clarity. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Reasoning

77. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - practical deployment framing - Variant 077

Objective: Pressure-test experimental design before expensive execution. Prioritize computational efficiency and produce publication-grade decisions for Control Groups.

Prompt: Audit the experimental design for Control Groups. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ICML reviewers. Include a stop/go decision protocol. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=skeptical statistician. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Rebuttal

78. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - method comparison clarity - Variant 078

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize clinical relevance and produce publication-grade decisions for Evaluation Protocol.

Prompt: Create a LaTeX writing workflow for Evaluation Protocol: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for AAAI submission checks. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=benchmarking specialist. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Validation

79. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - editorial risk reduction - Variant 079

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize publication readiness and produce publication-grade decisions for Bias Mitigation.

Prompt: Simulate a high-pressure Q&A; on Bias Mitigation. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=theory-oriented reviewer. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Optimization

80. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - venue-specific optimization - Variant 080

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize novelty and produce publication-grade decisions for Ablation Planning.

Prompt: Act as a senior research editor. Build a contribution architecture for Ablation Planning in multi-dataset benchmarking study targeting NeurIPS. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=methodology purist. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Planning

81. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - decision traceability - Variant 081

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize rigor and produce publication-grade decisions for Control Groups.

Prompt: Rewrite the methodology around Control Groups for replication package for supplementary material so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ICML. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=interdisciplinary committee member. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Execution

82. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - citation-strength positioning - Variant 082

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize reproducibility and produce publication-grade decisions for Evaluation Protocol.

Prompt: Write a Nature-style abstract for a study on Evaluation Protocol in high-variance real-world deployment. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=ethics compliance reviewer. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Compliance

83. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - defense-ready reasoning - Variant 083

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize clarity and produce publication-grade decisions for Bias Mitigation.

Prompt: Generate a figure strategy for Bias Mitigation with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by IEEE TPAMI. Add one caption rewrite for non-expert readers. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=reproducibility auditor. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Reasoning

84. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - high-confidence statistical reporting - Variant 084

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize statistical power and produce publication-grade decisions for Ablation Planning.

Prompt: Build an ethics checklist and mitigation plan for Ablation Planning in multi-dataset benchmarking study. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for NeurIPS review. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=applied domain expert. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Rebuttal

85. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - evidence-first writing - Variant 085

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize transferability and produce publication-grade decisions for Control Groups.

Prompt: Design a publication strategy for Control Groups: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ICML. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=systems performance reviewer. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Validation

86. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - hypothesis stress testing - Variant 086

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize ethical compliance and produce publication-grade decisions for Evaluation Protocol.

Prompt: Assume Reviewer #2 challenges Evaluation Protocol for AAAI. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile:

discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=industry impact reviewer. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Optimization

87. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - results-to-insight translation - Variant 087

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize computational efficiency and produce publication-grade decisions for Bias Mitigation.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Bias Mitigation. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for IEEE TPAMI. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=editor focused on clarity. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Planning

88. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - major-revision readiness - Variant 088

Objective: Select robust statistical strategy and interpretation logic. Prioritize clinical relevance and produce publication-grade decisions for Ablation Planning.

Prompt: Design a statistical analysis workflow for Ablation Planning with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for NeurIPS. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline};

resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=skeptical statistician. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Strategy

89. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - replication-grade documentation - Variant 089

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize publication readiness and produce publication-grade decisions for Control Groups.

Prompt: Analyze a draft on Control Groups and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ICML. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=benchmarking specialist. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Analysis

90. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - reviewer-resilient argumentation - Variant 090

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize novelty and produce publication-grade decisions for Evaluation Protocol.

Prompt: Interpret results for Evaluation Protocol using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in AAAI. Add one paragraph that preempts overclaiming. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=theory-oriented reviewer. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Reasoning

91. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - ethics-by-design reporting - Variant 091

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize rigor and produce publication-grade decisions for Bias Mitigation.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Bias Mitigation and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for IEEE TPAMI. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=methodology purist. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Rebuttal

92. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - limitations-first transparency - Variant 092

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize reproducibility and produce publication-grade decisions for Ablation Planning.

Prompt: Generate a publication-ready section for Ablation Planning tuned for NeurIPS. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=interdisciplinary committee member. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Validation

93. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - cross-domain readability - Variant 093

Objective: Pressure-test experimental design before expensive execution. Prioritize clarity and produce publication-grade decisions for Control Groups.

Prompt: Audit the experimental design for Control Groups. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ICML reviewers. Include a stop/go decision protocol. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=ethics compliance reviewer. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Optimization

94. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - impact-focused communication - Variant 094

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize statistical power and produce publication-grade decisions for Evaluation Protocol.

Prompt: Create a LaTeX writing workflow for Evaluation Protocol: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for AAAI submission checks. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=reproducibility auditor. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Planning

95. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - publishable novelty framing - Variant 095

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize transferability and produce publication-grade decisions for Bias Mitigation.

Prompt: Simulate a high-pressure Q&A; on Bias Mitigation. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=applied domain expert. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Strategy

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96. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - practical deployment framing - Variant 096

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize ethical compliance and produce publication-grade decisions for Ablation Planning.

Prompt: Act as a senior research editor. Build a contribution architecture for Ablation Planning in multi-dataset benchmarking study targeting NeurIPS. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=systems performance reviewer. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Analysis

97. Build Reviewer-Proof Ablations: power-aware experiment planning for replication package for supplementary material (ICML) - method comparison clarity - Variant 097

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize computational efficiency and produce publication-grade decisions for Control Groups.

Prompt: Rewrite the methodology around Control Groups for replication package for supplementary material so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ICML. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ICML; reviewer persona=industry impact reviewer. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Control Groups.

Difficulty: Beginner

Category: Writing

98. Create a Robust Validation Plan: evaluation matrix design for high-variance real-world deployment (AAAI) - editorial risk reduction - Variant 098

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize clinical relevance and produce publication-grade decisions for Evaluation Protocol.

Prompt: Write a Nature-style abstract for a study on Evaluation Protocol in high-variance real-world deployment. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=AAAI; reviewer persona=editor focused on clarity. Example outcome: decision log with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready decision log that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Evaluation Protocol.

Difficulty: Intermediate

Category: Publication

99. Design Controls That Survive Peer Review: confound-control workflow for resource-constrained lab experiment (IEEE TPAMI) - venue-specific optimization - Variant 099

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize publication readiness and produce publication-grade decisions for Bias Mitigation.

Prompt: Generate a figure strategy for Bias Mitigation with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by IEEE TPAMI. Add one caption rewrite for non-expert readers. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE TPAMI; reviewer persona=skeptical statistician. Example outcome: experiment protocol with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready experiment protocol that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Bias Mitigation.

Difficulty: Advanced

Category: Validation

100. Strengthen Experimental Design: ablation architecture for multi-dataset benchmarking study (NeurIPS) - decision traceability - Variant 100

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize novelty and produce publication-grade decisions for Ablation Planning.

Prompt: Build an ethics checklist and mitigation plan for Ablation Planning in multi-dataset benchmarking study. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for NeurIPS review. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=NeurIPS; reviewer persona=benchmarking specialist. Example outcome: variable registry with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready variable registry that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Ablation Planning.

Difficulty: Expert

Category: Optimization

Summary

Applying these 100 workflows systematically improves decision quality, writing clarity, and publication readiness across research cycles.